```
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# div : B
# Roll No :31
# PRN : 72332154L
# DMV PRATICAL NO 5
```

```
import kagglehub
import pandas as pd
import matplotlib.pyplot as plt
import os

print("Libraries imported successfully!")
```

(module) pandas

```
Libraries imported successfully!
```

```
# Download Air Quality dataset
path = kagglehub.dataset_download("rohanrao/air-quality-data-in-india")

print("Dataset downloaded successfully!")
print("Dataset path:", path)
print("Files available:", os.listdir(path))
```

```
Using Colab cache for faster access to the 'air-quality-data-in-india' dataset.
Dataset downloaded successfully!
Dataset path: /kaggle/input/air-quality-data-in-india
Files available: ['stations.csv', 'station_hour.csv', 'city_day.csv', 'city_hour.csv', 'station_day.csv']
```

```
# Load the CSV file
file_path = os.path.join(path, "city_day.csv")

df = pd.read_csv(file_path)

# Display first 5 rows
df.head()
```

	City	Date	PM2.5	PM10	NO	NO2	NOx	NH3	CO	SO2	O3	Benzene	Toluene	Xylene	AQI	AQI_Bucket
0	Ahmedabad	2015-01-01	NaN	NaN	0.92	18.22	17.15	NaN	0.92	27.64	133.36	0.00	0.02	0.00	NaN	NaN
1	Ahmedabad	2015-01-02	NaN	NaN	0.97	15.69	16.46	NaN	0.97	24.55	34.06	3.68	5.50	3.77	NaN	NaN
2	Ahmedabad	2015-01-03	NaN	NaN	17.40	19.30	29.70	NaN	17.40	29.07	30.70	6.80	16.40	2.25	NaN	NaN

Next steps: Generate code with df | New interactive sheet

```
# Check number of rows and columns
print("Dataset Shape:", df.shape)

# Number of rows
print("Number of rows:", df.shape[0])

# Number of columns
print("Number of columns:", df.shape[1])
```

```
Dataset Shape: (29531, 16)
Number of rows: 29531
Number of columns: 16
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 29531 entries, 0 to 29530
Data columns (total 16 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   City        29531 non-null  object
 1   Date        29531 non-null  object
 2   PM2.5       24933 non-null  float64
 3   PM10        18391 non-null  float64
 4   NO          25949 non-null  float64
 5   NO2         25946 non-null  float64
 6   NOx         25346 non-null  float64
 7   NH3         19203 non-null  float64
 8   CO          27472 non-null  float64
 9   SO2         25677 non-null  float64
 10  O3          25509 non-null  float64
```

```
 11  Benzene     23908 non-null  float64
 12  Toluene     21490 non-null  float64
 13  Xylene      11422 non-null  float64
 14  AQI         24850 non-null  float64
 15  AQI_Bucket  24850 non-null  object
dtypes: float64(13), object(3)
memory usage: 3.6+ MB
```

```
df.describe()
```

	PM2.5	PM10	NO	NO2	NOx	NH3	CO	SO2	
count	24933.000000	18391.000000	25949.000000	25946.000000	25346.000000	19203.000000	27472.000000	25677.000000	25509.000
mean	67.450578	118.127103 (module) pandas	17.574730	28.560659	32.309123	23.483476	2.248598	14.531977	34.491
std	64.661449	90.605110	22.785846	24.474746	31.646011	25.684275	6.962884	18.133775	21.694
min	0.040000	0.010000	0.020000	0.010000	0.000000	0.010000	0.000000	0.010000	0.010
25%	28.820000	56.255000	5.630000	11.750000	12.820000	8.580000	0.510000	5.670000	18.860
50%	48.570000	95.680000	9.890000	21.690000	23.520000	15.850000	0.890000	9.160000	30.840
75%	80.590000	149.745000	19.950000	37.620000	40.127500	30.020000	1.450000	15.220000	45.570
max	949.990000	1000.000000	390.680000	362.210000	467.630000	352.890000	175.810000	193.860000	257.730

```
city_data = df[df["City"] == "Mumbai"].copy()

print("Mumbai Air Quality Data:")
display(city_data.head())
```

```
Mumbai Air Quality Data:
```

	City	Date	PM2.5	PM10	NO	NO2	NOx	NH3	CO	SO2	O3	Benzene	Toluene	Xylene	AQI	AQI_Bucket
21855	Mumbai	2015-01-01	NaN	NaN	NaN	NaN	27.38	NaN	0.0	NaN	NaN	0.0	0.0	0.0	NaN	NaN
21856	Mumbai	2015-01-02	NaN	NaN	NaN	NaN	26.20	NaN	0.0	NaN	NaN	0.0	0.0	0.0	NaN	NaN
21857	Mumbai	2015-01-03	NaN	NaN	NaN	NaN	26.04	NaN	0.0	NaN	NaN	0.0	0.0	0.0	NaN	NaN

```
city_data["Date"] = pd.to_datetime(city_data["Date"])

city_data = city_data.sort_values("Date")

display(city_data.head())
```

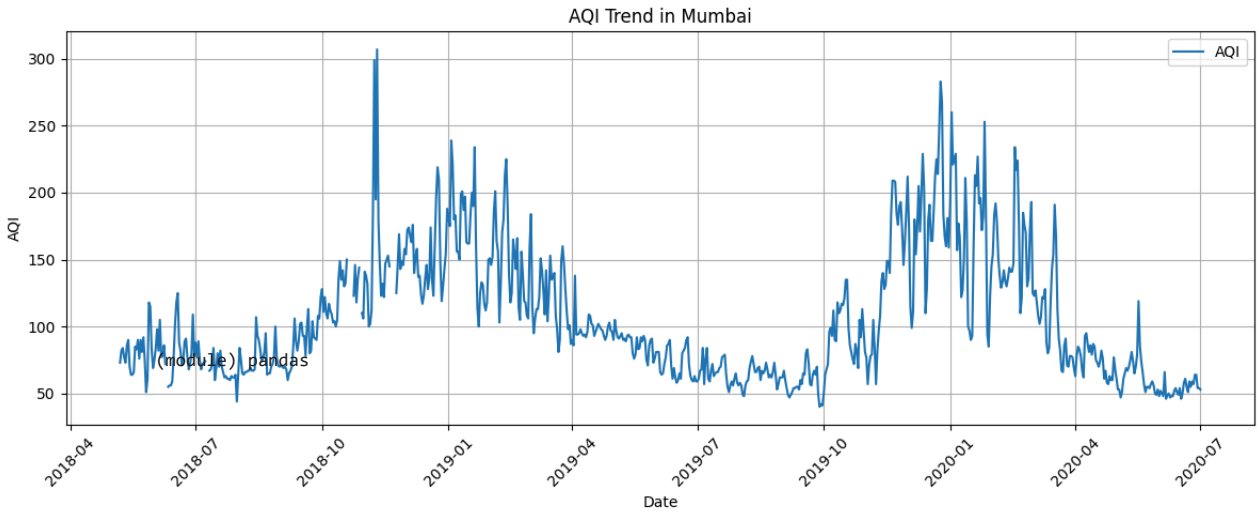
	City	Date	PM2.5	PM10	NO	NO2	NOx	NH3	CO	SO2	O3	Benzene	Toluene	Xylene	AQI	AQI_Bucket
21855	Mumbai	2015-01-01	NaN	NaN	NaN	NaN	27.38	NaN	0.0	NaN	NaN	0.0	0.0	0.0	NaN	NaN
21856	Mumbai	2015-01-02	NaN	NaN	NaN	NaN	26.20	NaN	0.0	NaN	NaN	0.0	0.0	0.0	NaN	NaN
21857	Mumbai	2015-01-03	NaN	NaN	NaN	NaN	26.04	NaN	0.0	NaN	NaN	0.0	0.0	0.0	NaN	NaN

```
plt.figure(figsize=(12, 5))

plt.plot(city_data["Date"], city_data["AQI"], label="AQI")

plt.xlabel("Date")
plt.ylabel("AQI")
plt.title("AQI Trend in Mumbai")
plt.legend()
plt.grid(True)
plt.xticks(rotation=45)

plt.tight_layout()
plt.show()
```

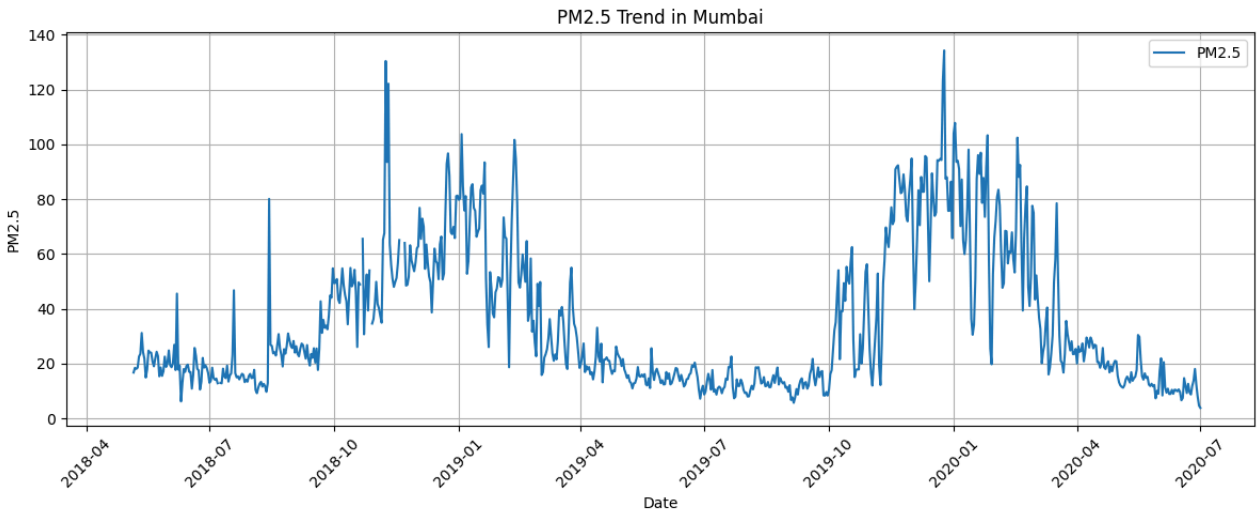



```
plt.figure(figsize=(12, 5))

plt.plot(city_data["Date"], city_data["PM2.5"], label="PM2.5")

plt.xlabel("Date")
plt.ylabel("PM2.5")
plt.title("PM2.5 Trend in Mumbai")
plt.legend()
plt.grid(True)
plt.xticks(rotation=45)

plt.tight_layout()
plt.show()
```

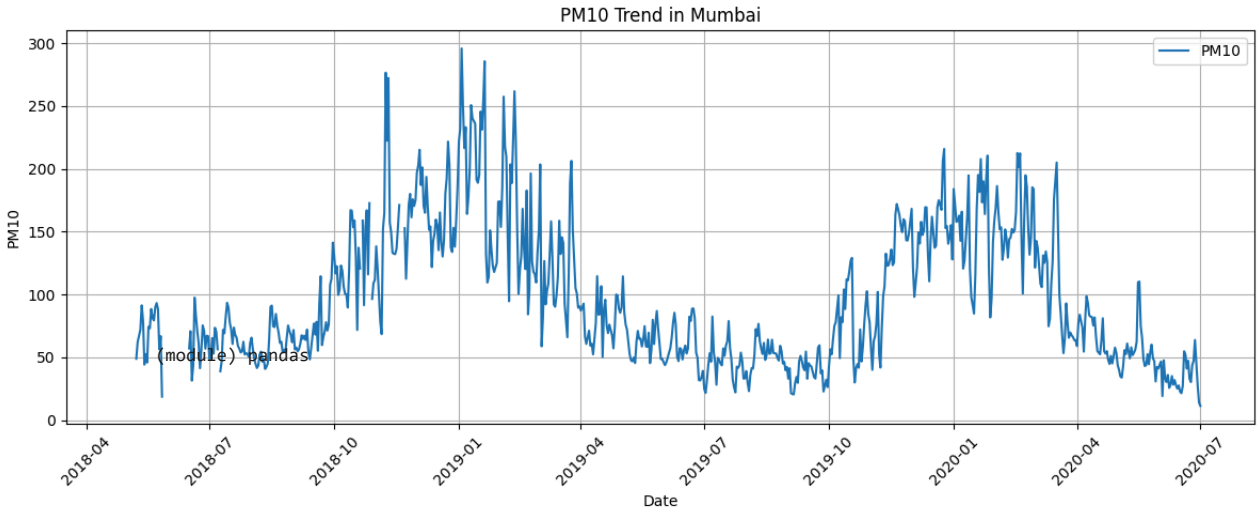



```
plt.figure(figsize=(12, 5))

plt.plot(city_data["Date"], city_data["PM10"], label="PM10")

plt.xlabel("Date")
plt.ylabel("PM10")
plt.title("PM10 Trend in Mumbai")
plt.legend()
plt.grid(True)
plt.xticks(rotation=45)

plt.tight_layout()
plt.show()
```

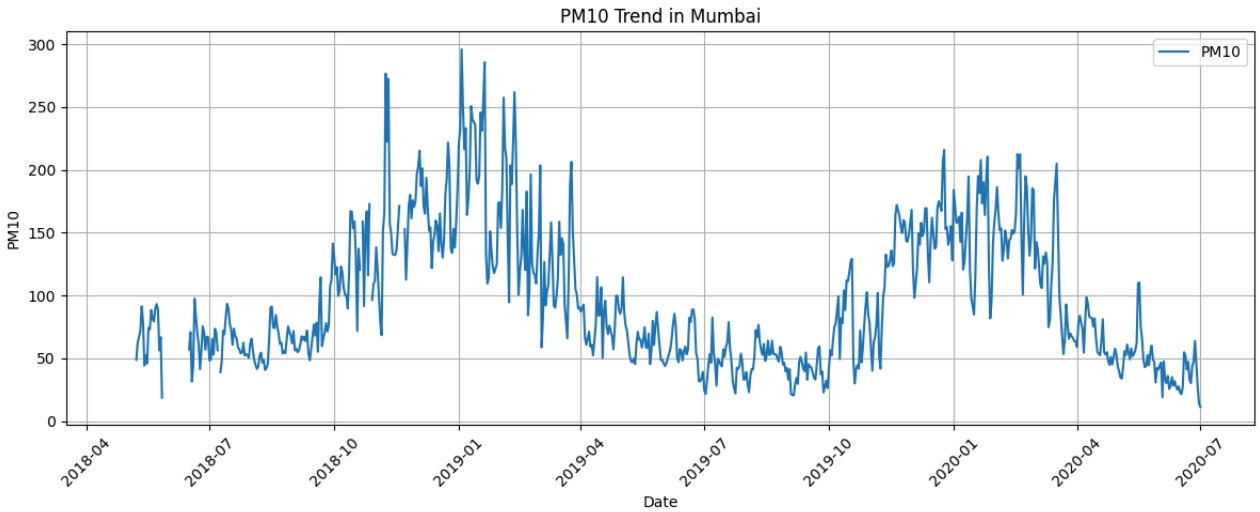



```
plt.figure(figsize=(12, 5))

plt.plot(city_data["Date"], city_data["PM10"], label="PM10")

plt.xlabel("Date")
plt.ylabel("PM10")
plt.title("PM10 Trend in Mumbai")
plt.legend()
plt.grid(True)
plt.xticks(rotation=45)

plt.tight_layout()
plt.show()
```

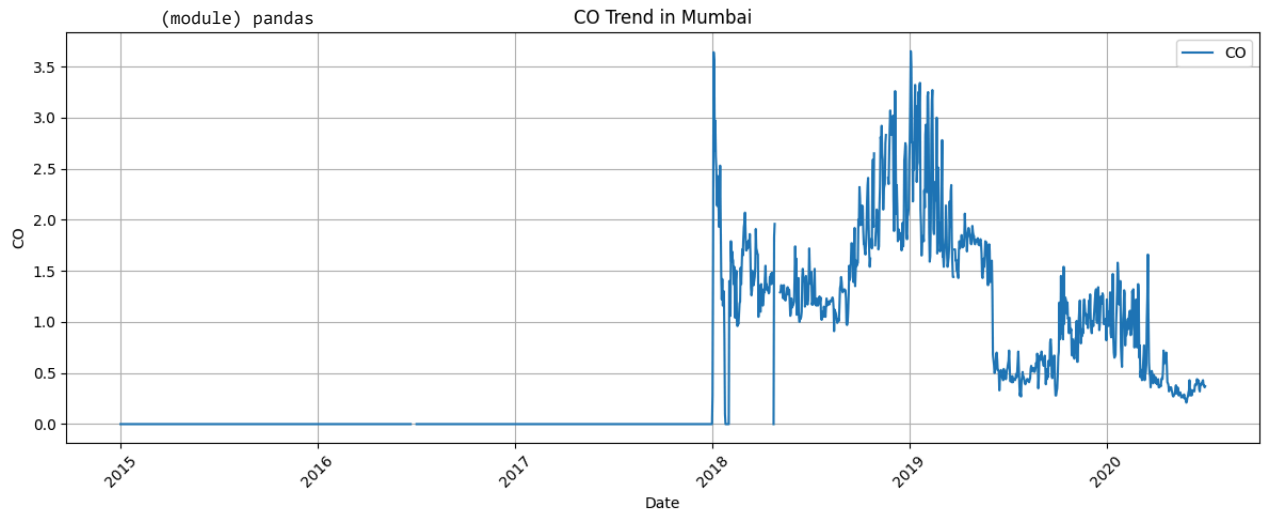


```
plt.figure(figsize=(12, 5))

plt.plot(city_data["Date"], city_data["CO"], label="CO")

plt.xlabel("Date")
plt.ylabel("CO")
plt.title("CO Trend in Mumbai")
plt.legend()
plt.grid(True)
plt.xticks(rotation=45)

plt.tight_layout()
plt.show()
```



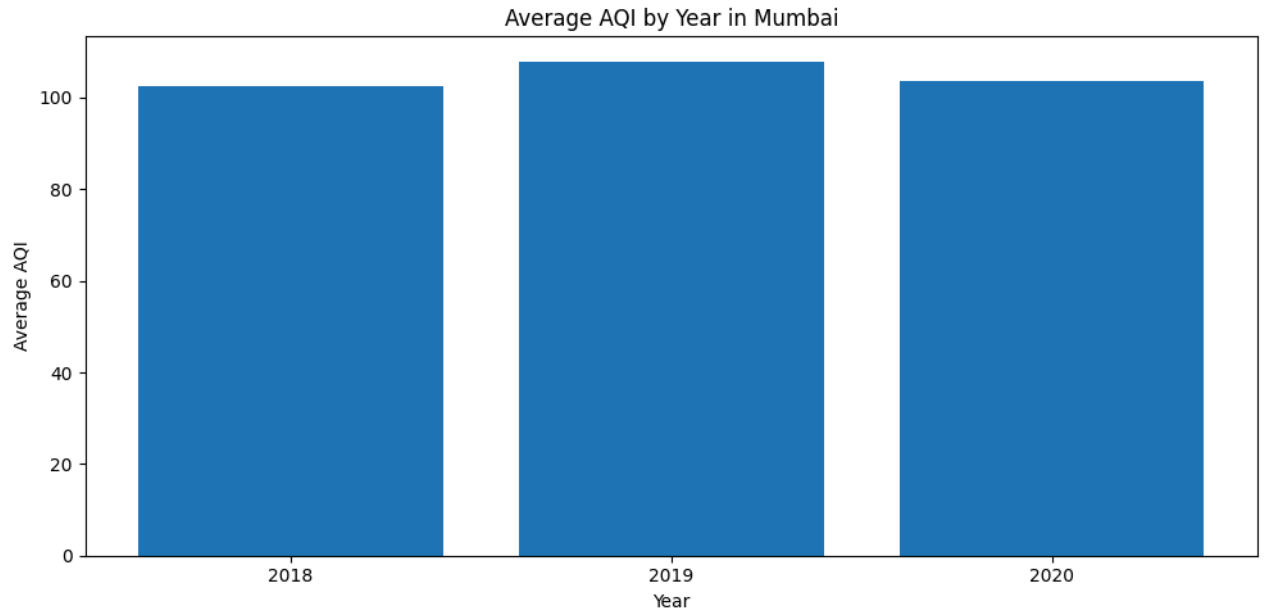

```
yearly_aqi = city_data.groupby(
    city_data["Date"].dt.year
)["AQI"].mean()

plt.figure(figsize=(10, 5))

plt.bar(yearly_aqi.index.astype(str), yearly_aqi.values)

plt.xlabel("Year")
plt.ylabel("Average AQI")
plt.title("Average AQI by Year in Mumbai")

plt.tight_layout()
plt.show()
```



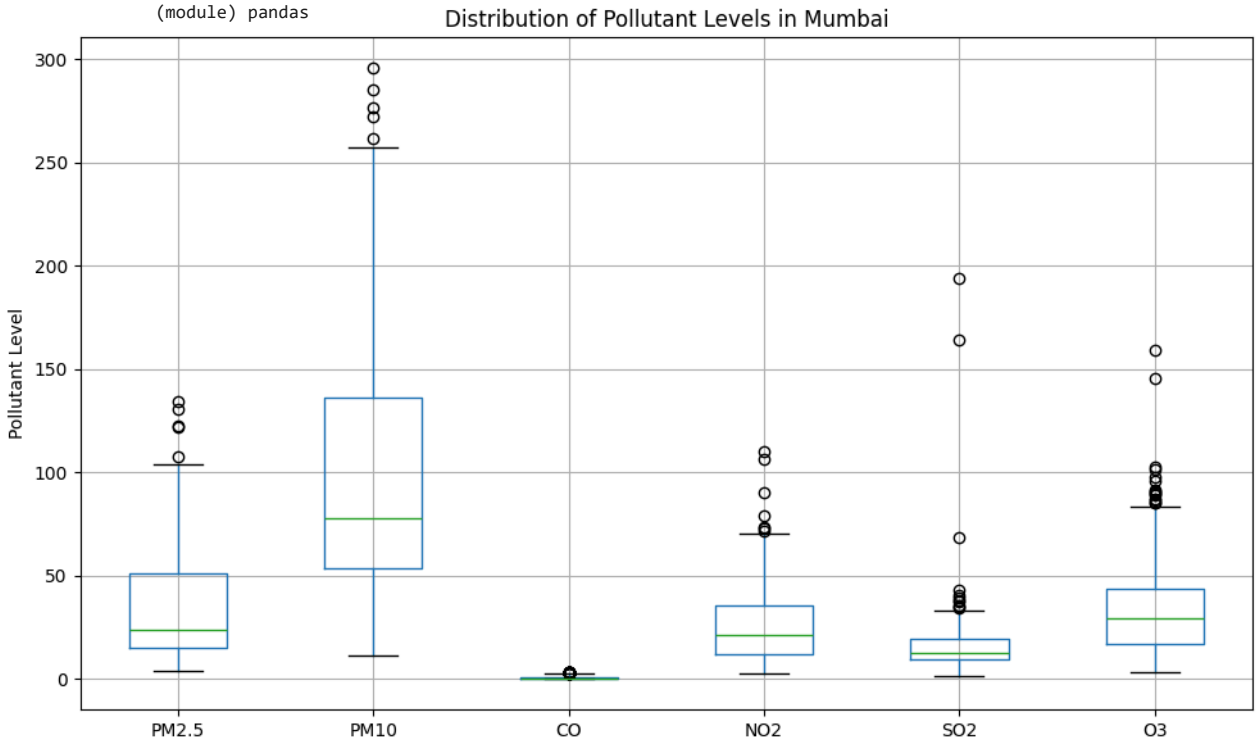
```
plt.figure(figsize=(10, 6))

city_data[
    ["PM2.5", "PM10", "CO", "NO2", "SO2", "O3"]
].boxplot()

plt.title("Distribution of Pollutant Levels in Mumbai")
plt.ylabel("Pollutant Level")
plt.grid(True)

plt.tight_layout()
plt.show()
```

(module) pandas

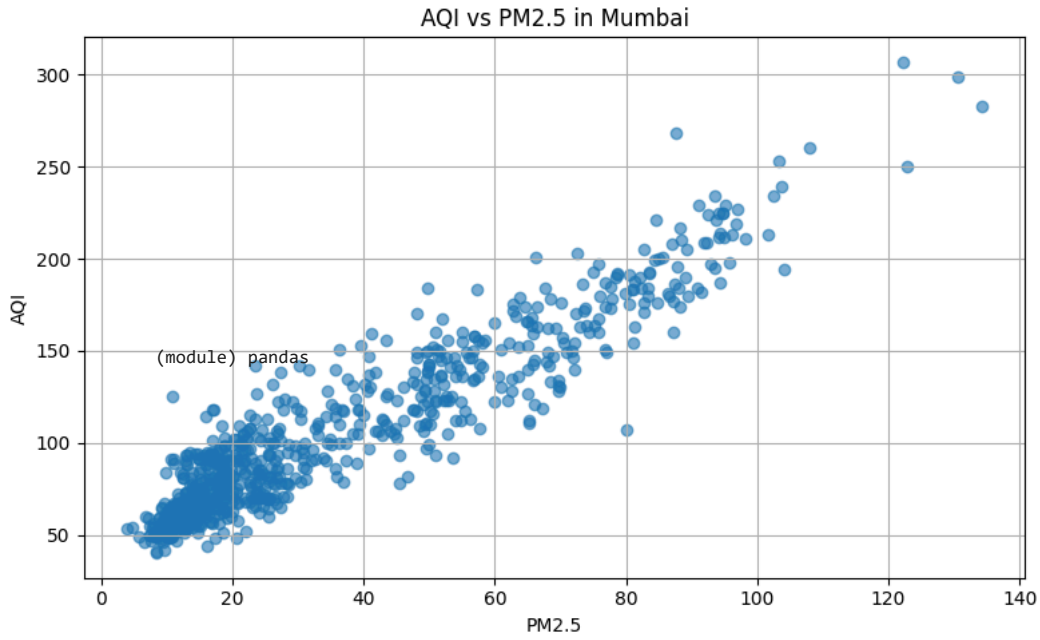



```
plt.figure(figsize=(8, 5))

plt.scatter(
    city_data["PM2.5"],
    city_data["AQI"],
    alpha=0.6
)

plt.xlabel("PM2.5")
plt.ylabel("AQI")
plt.title("AQI vs PM2.5 in Mumbai")
plt.grid(True)

plt.tight_layout()
plt.show()
```

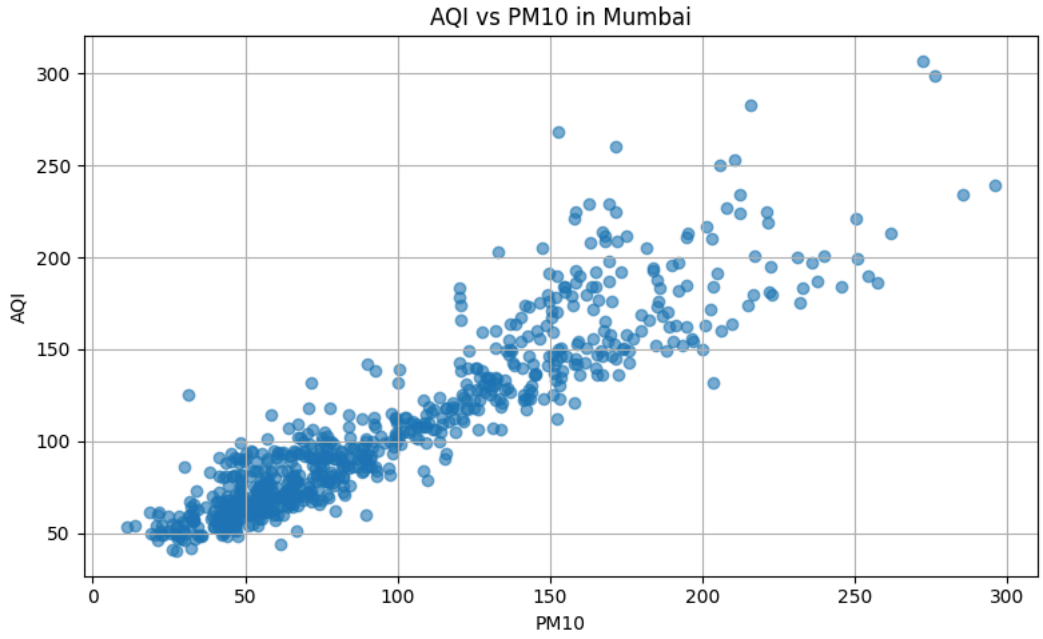



```
plt.figure(figsize=(8, 5))

plt.scatter(
    city_data["PM10"],
    city_data["AQI"],
    alpha=0.6
)

plt.xlabel("PM10")
plt.ylabel("AQI")
plt.title("AQI vs PM10 in Mumbai")
plt.grid(True)

plt.tight_layout()
plt.show()
```



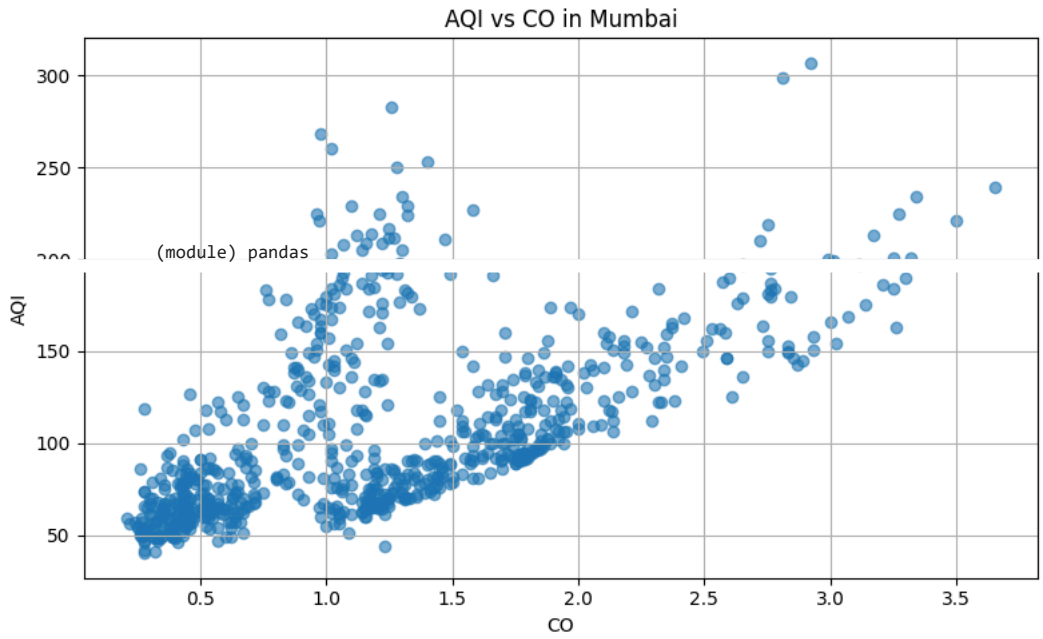

```
plt.figure(figsize=(8, 5))

plt.scatter(
    city_data["CO"],
    city_data["AQI"],
    alpha=0.6
)

plt.xlabel("CO")
plt.ylabel("AQI")
plt.title("AQI vs CO in Mumbai")
```

```
plt.grid(True)

plt.tight_layout()
plt.show()
```




```
plt.figure(figsize=(12, 6))

plt.plot(
    city_data["Date"],
    city_data["PM2.5"],
    label="PM2.5"
)

plt.plot(
    city_data["Date"],
    city_data["PM10"],
    label="PM10"
)

plt.plot(
    city_data["Date"],
    city_data["CO"],
    label="CO"
)

plt.xlabel("Date")
plt.ylabel("Pollutant Level")
plt.title("Pollutant Trends in Mumbai")
plt.legend()
plt.grid(True)
plt.xticks(rotation=45)

plt.tight_layout()
plt.show()
```

